LUKE THOMPSON

ASSOCIATE EMBEDDED SOFTWARE DEVELOPER

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Portfolio: xylardark.github.io/Portfolio-Studio/u/admin/l3harris · GitHub: github.com/XylarDark

PROFESSIONAL SUMMARY

Embedded software developer with C++ on Arm microcontrollers and automated test tooling. At TruckSail (Feb 2020 – Jan 2021) I wrote C++ firmware for truck ECU devices (STM32, CAN bus) and did defect finding for reliability. Since February 2026 I have been writing C++ and Python test and build automation on HomeWorld, my solo Unreal Engine 5.8 project. Earlier: game UI/UX and C# REST integrations (OutFusion) and client software and QA (Evenica).

CORE SKILLS

Languages	C++, Python, TypeScript/JavaScript, C#, SQL, HTML/CSS
Embedded	STM32 (Arm), CAN bus, firmware defect finding
Testing & tooling	Automated test runners, pytest, Vitest, Git/GitHub, build automation
Hosts	Linux and Windows

SELECTED PROJECTS

HomeWorld — C++ game, test and build automation Feb 2026 – present

Independent (solo) · Unreal Engine 5.8, C++, Python · private repository, available on request

- Write C++ gameplay and engine-side code, and Python tooling that drives the editor and checks results against real state.
- Built a headless test runner that hard-fails on five causes of fake passes (unreadable report, wrong engine, zero tests run, stale report, stale binaries), each with a regression test.
- Built a build wrapper with one machine-wide queue so parallel jobs cannot start colliding builds; 42 C++ automation test files and 150+ Python test files.
- Some of this code is written by AI coding agents under my direction; source changes wait for my review before they merge.

Portfolio Studio — media-résumé site Sep 2026 – present

Independent · React, TypeScript, Vite, Supabase · xylardark.github.io/Portfolio-Studio

- React/TypeScript portfolio product with per-employer pages, built through pull requests I review; headless-Chrome checks at desktop and phone sizes, and GitHub Actions deploys each merge.

Shift — playable shop & delivery game Aug – Sep 2026

Independent · Phaser 3, TypeScript, Vite · Live: xylardark.github.io/ShiftGame

- Shipped a TypeScript browser/PWA game with a fixed 60 Hz simulation; 83 Vitest test files in the public repo.

EXPERIENCE

Game Developer · OutFusion · Hamilton, ON Feb 2022 – Dec 2022

- Implemented game UI/UX from the design docs; built and tested C# REST API integrations for player data; integrated and tested game UI across mobile and desktop; partnered on bug verification and release quality.

Software Developer · TruckSail · Ancaster, ON Feb 2020 – Jan 2021

- Wrote C++ firmware for truck ECU devices (STM32, CAN bus) with defect finding for reliability; built JS/Bootstrap UIs and optimized SQL; maintained GitHub workflows and did cross-browser QA.

Software Developer & QA Tester · Evenica · Burlington, ON 2018 – Sep 2019

- Built and tested client software (Andrew Peller, Young's Market Company, MM Realty International): T-SQL, C#, Linux, and Laravel (MySQL, PHP, JavaScript/jQuery, Vue.js).

Delivery Driver (Uber, retail delivery) · Hamilton, ON 2022 – 2026

EDUCATION

Diploma, Computer Systems Technology — Software Development · Mohawk College · Hamilton, ON · 2019

Valid Ontario Class G licence · Open to on-site work in Waterdown (commuting from Hamilton)